

1.8 Exponentials & Logarithms

Mathematics B (MEI) — OCR B — A-Level (H640) — Pure Mathematics — spec refs E1–E11

What this topic covers. Exponential and logarithmic functions and their graphs, the laws of logarithms, the special role of e and $\ln$, reducing power and exponential laws to linear form, and exponential growth and decay models together with their limitations.

1. Exponential functions

For $a > 0$, the graph of $y = a^x$ passes through $(0, 1)$, is always positive, and has the x -axis as a horizontal asymptote (spec ref E1).

- $a > 1$: increasing (growth).
- $0 < a < 1$: decreasing (decay).

2. Logarithms as inverses

$\log_a x$ is the **inverse** of a^x :

$$x = a^y \iff y = \log_a x \quad (a > 0, x > 0)$$

Because they are inverses, the graph of $y = \log_a x$ is the reflection of $y = a^x$ in $y = x$: it passes through $(1, 0)$, is defined only for $x > 0$, and has the y -axis as a **vertical** asymptote (E3).

Converting between forms. $2^5 = 32$ becomes $\log_2 32 = 5$. $\log_3 9 = 2$ becomes $3^2 = 9$.

3. Laws of logarithms

$$\log_a(xy) = \log_a x + \log_a y$$

$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$

$$\log_a(x^k) = k \log_a x$$

Two values follow immediately (E5):

$$\log_a a = 1, \quad \log_a 1 = 0$$

The power law covers $k = -1$ and $k = \frac{1}{2}$, so $\log_a\left(\frac{1}{x}\right) = -\log_a x$ and $\log_a \sqrt{x} = \frac{1}{2} \log_a x$.

Change of base

$$\log_a b = \frac{\log b}{\log a}$$

There is no law for $\log(x + y)$. Writing $\log(x + y) = \log x + \log y$ is wrong — the addition law applies to a **product** inside the logarithm, not a sum.

4. Solving exponential equations

Take logarithms of both sides, then use the power law to bring the unknown down (E6).

Worked example. Solve $3^x = 20$.

$$x \log 3 = \log 20 \implies x = \frac{\log 20}{\log 3} = 2.727 \text{ (3 d.p.)}$$

Worked example. Solve $\log_2 x + \log_2(x - 2) = 3$.

$$\log_2(x(x - 2)) = 3 \implies x^2 - 2x = 8 \implies (x - 4)(x + 2) = 0$$

So $x = 4$ or $x = -2$. But $\log_2(x - 2)$ requires $x > 2$, so $x = -2$ is rejected: $x = 4$.

Always check solutions against the domain of the logarithm. A logarithm of a negative number or zero is undefined, and a rejected root that is left in costs the final mark.

Worked example. Solve $e^x - 5e^{-x} = 4$.

Let $u = e^x$. Multiplying through by u :

$$u^2 - 4u - 5 = 0 \implies (u - 5)(u + 1) = 0$$

$u = e^x$ is always positive, so $u = -1$ is impossible. Then $e^x = 5$, giving $x = \ln 5$.

5. e and the natural logarithm

$e \approx 2.718$ is the base for which the gradient of $y = e^x$ equals e^x itself. More generally the gradient of e^{kx} is ke^{kx} (E9).

That property is exactly why the exponential model suits so many situations: whenever a quantity grows at a rate proportional to its current size, the solution is an exponential.

$\ln x$ means $\log_e x$, the inverse of e^x (E10). So $\ln e = 1$, $\ln 1 = 0$, and:

$$e^{\ln x} = x, \quad \ln(e^x) = x$$

6. Reducing to linear form

Two standard laws can be turned into straight lines by taking logarithms, so that experimental data can be used to estimate the parameters (E7).

Law	Take logs to get	Plot	Gradient	Intercept
$y = ax^n$	$\log y = \log a + n \log x$	$\log y$ against $\log x$	n	$\log a$
$y = ab^x$	$\log y = \log a + x \log b$	$\log y$ against x	$\log b$	$\log a$

The distinction is which axis carries a logarithm. A **power** law needs log–log; an **exponential** law needs log y against plain x . If a log–log plot gives a straight line, the relationship is a power law.

Worked example. Data plotted as log y against log x give a straight line with gradient 1.5 and vertical intercept 0.6. Find the law.

The form is $y = ax^n$ with $n = 1.5$ and $\log a = 0.6$, so $a = 10^{0.6} \approx 3.98$, giving $y \approx 3.98 x^{1.5}$.

7. Growth and decay models

The standard models (E11):

$$\text{growth: } N = N_0 e^{kt}, \quad \text{decay: } N = N_0 e^{-kt} \quad (k > 0)$$

N_0 is the initial value, at $t = 0$.

Worked example. A substance decays as $N = N_0 e^{-kt}$ with half-life 5. Find k .

At $t = 5$ the amount is halved:

$$e^{-5k} = \frac{1}{2} \implies -5k = \ln \frac{1}{2} = -\ln 2 \implies k = \frac{\ln 2}{5}$$

Worked example. For $y = 5e^{-0.2t}$, state the initial value and the long-term value.

At $t = 0$, $y = 5$. As $t \rightarrow \infty$, $e^{-0.2t} \rightarrow 0$, so $y \rightarrow 0$.

Other contexts named in the specification include continuous compound interest, Pe^{rt} , and drug concentration decay.

Limitations and refinements

MEI explicitly asks for **criticism** of these models. Standard points: unlimited exponential growth ignores finite resources, so a population model needs a carrying capacity; a decay model tending to

zero may be unrealistic if some background level persists; and the constant k may not stay constant over long periods.

8. Common pitfalls

- Inventing a law for $\log(x + y)$.
- Failing to reject a root that makes a logarithm undefined.
- Forgetting that e^x is always positive when solving a disguised quadratic.
- Confusing which quantities to plot for a power law versus an exponential law.
- Reading the intercept of a log-linear plot as a rather than $\log a$.
- Writing the asymptote of $y = \ln x$ as $y = 0$ — it is $x = 0$.
- Losing the minus sign in a decay exponent.